

Homework
Due: 05.13
Noisy signals filtered by low pass filters

Please turn in the Matlab simulation codes and results.

- Matlab simulations 1 (analog filter)
 - Create the sampled data for the following signal: $0.5 \sin(2\pi ft) + \text{noise}(t)$, where noise is zero mean, standard deviation of 2, $f = 0.3$ Hz, sampling frequency 1 kHz, data length of 10 sec.
 - Use Matlab function to create 3 “analog”, Butterworth low-pass filter with the passband of 1 Hz, 10 Hz, and 100 Hz, respectively.
 - Use the Matlab function “lsim” to simulate the output of filtered signal. Plot the time domain signal of the original signal, 1 Hz low-pass filtered signal, 10 Hz low-pass filtered signal, and 100 Hz low-pass filtered signal.
 - Use “Fourier transform” to find the frequency content of the signal. Plot the frequency content for the original signal, 1 Hz low-pass filtered signal, 10 Hz low-pass filtered signal, and 100 Hz low-pass filtered signal.
 - Matlab functions: randn, butter, std, mean, fft, lsim, etc.

- Matlab simulations 2 (digital filter)
 - Create an input signal: $x(N) = 2.5 \sin(2\pi \cdot 5 \cdot N \cdot dt) + 10 \sin(2\pi \cdot 100 \cdot N \cdot dt) + n(N \cdot dt)$, where $n(t)$ is a random signal with zero mean and the standard deviation of 2; dt is the sampling time of 1ms.
 - Use Matlab to design two “digital” low pass filters, where the first one is 1st order butterworth filter, the second one is 2nd order butterworth filter, and the 3rd one is the 2nd order Chebyshev filter. Three filters are with pass band of 10Hz
 - Plot the time-response and FFT results of the filtered signal.
 - Matlab functions: randn, butter, cheby1, std, mean, filter, fft, etc.